

System Proposal & Comprehensive Scope of Work

A Unified Multi-Tier Digital Architecture for Luxury Safari Car Rental, Verified Fleet Host Operations, Concurrency Trip Locking, and Telematics

Executive Purpose: This document establishes the technical blueprint, functional scope, operational rules, and architectural deliverables for the **M-TRAVEL Tours Digital Platform**. The platform unifies luxury vehicle rental, safari game drive expeditions, long-distance bus ticketing, and holiday home reservations across Kenya into a secure, real-time operating ecosystem.

Key architectural pillars include **strict multi-tenant account privacy** (zero cross-account data leakage), **Admin-exclusive live vehicle activation**, **real-time concurrency trip locking**, and automated financial settlements via Safaricom Lipa Na M-Pesa STK Push and B2C Host Disburse.

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Prepared For: M-Travel Steering Committee & Board

Lead Engineering: MonarchKingz Architecture Team

Target Markets: Nairobi, Maasai Mara, Diani, Amboseli

Classification: Commercial in Confidence

1. Executive Summary & Market Context

Kenya's private transport and safari tourism sector generates substantial annual revenue across domestic and international travelers. However, traditional vehicle hiring remains plagued by manual phone/WhatsApp coordination, double-booking risks, unverified vehicle quality, and zero real-time vehicle telemetry for vehicle owners. **M-TRAVEL TOURS** eliminates these inefficiencies through a centralized digital marketplace that delivers real-time live availability, automated booking locks, and bank-grade data security.

Industry Vulnerabilities	The M-TRAVEL Architectural Solution
- Manual WhatsApp booking results in double-booking conflicts. - Fleet owners have zero telemetry once cars depart on trips. - High dispute rate surrounding cash deposits and delayed payouts. - Unverified hosts risk tourist safety and damage brand credibility. - Notifications and alerts leak across unsegregated user accounts.	Real-Time Marketplace Sync: Automatic vehicle lock on booking confirmation. Live Telematics & GPS: Integrated telemetry tracking routes across Kenya. Daraja M-Pesa Rails: 1-Click STK push & instant B2C host disbursements. Admin Quality Gate: Cars start offline and go live ONLY after Admin approval. Strict Data Isolation: Cryptographic separation ensures zero cross-host data leaks.

2. Enterprise System Architecture & Privacy Standard

Tier	Technology Stack	Key Functional Deliverables
Client Presentation	React 18, Vite, TypeScript, Tailwind CSS, Redux Toolkit	Fast responsive UI, 3D interactive viewer, cross-tab reactive state sync.
Microservices API	NestJS, TypeScript, Node.js v24, Supabase Edge Functions	State validation, reservation orchestration, Daraja webhook handlers.
Database & Sync	PostgreSQL 15 with RLS, Supabase Realtime Channels	Immutable transactional ledger, realtime pub/sub event bus, encrypted credentials.
FinTech Gateway	Safaricom Daraja 2.0 API (STK Push, C2B, B2C Disburse)	Instant STK push checkout, automated callback ledgering, host wallet payouts.
Telematics Engine	OpenCV vehicle telemetry simulator, Leaflet / Mapbox GIS	Live driver coordinates, trip telemetry, route monitoring to national parks.

STRICT MULTI-TENANT PRIVACY ARCHITECTURE (Guaranteed Zero Data Leakage):

- Fleet Host Isolation:** A Fleet Host can strictly only access cars registered under their unique account, bookings made for their vehicles, their net earnings, and notifications addressed directly to their user ID. Other hosts' inventories and financial performance are completely shielded.
- Tourist Isolation:** Tourists only access their personal bookings, GPS tracking sessions, and receipts.
- Notification Engine:** All alerts enforce strict `recipient_id === userId` matching; cross-user notification leakage is completely eliminated.

3. Detailed Functional Scope of Work

Pillar 1: Traveler & Tourist Booking Portal

- **Catalog & Smart Search:** Multi-category filtering across 4x4 Safari Land Cruisers, Executive Vans, Luxury Sedans, Buses, and Holiday Homes with live currency toggle (KES / USD).
- **Expedition Customization:** Add-on options for certified professional safari guides/chauffeurs and comprehensive collision insurance seamlessly computed into trip totals.
- **Lipa Na M-Pesa STK Push Checkout:** 1-Click checkout triggering an automated PIN authorization prompt on the user's phone, verified via Daraja callback with instant confirmation.
- **Tourist Dashboard & Telematics:** Real-time trip status tracking, driver details, interactive GPS vehicle tracker, trip health scoring, and downloadable official digital receipts.
- **Strict Tourist Privacy:** Scoped data visibility preventing tourists from accessing other travelers' reservations.

Pillar 2: Fleet Host (Vehicle Owner) Operations Portal

- **Clean Host Workspace:** Dedicated operational portal that eliminates consumer browsing elements, focusing entirely on vehicle fleet oversight, incoming bookings, earnings, and M-Pesa payouts.
- **Smart Vehicle Registration Wizard:** Multi-step vehicle onboarding featuring client-side canvas photo compression (front, rear, interior), specification registration, and automated document submission.
- **Default Offline & Pending State:** All newly registered cars default strictly to PENDING_APPROVAL with isLive: false, preventing unverified vehicles from reaching tourists.
- **Read-Only Marketplace Status Display:** Hosts cannot manually toggle cars live. Instead, they receive clear visual indicators: 'Awaiting Admin Approval', 'Admin Verified Live', or 'Auto-Locked (On Active Trip)'.
- **Reservation Management:** Real-time acceptance/rejection of booking requests and trip completion confirmation.
- **Automated Financial Reconciliation:** Instant 85% revenue credit to the host's wallet upon trip completion (15% platform commission deducted automatically), with instant M-Pesa B2C withdrawal.

Pillar 3: Platform Administration & Quality Control (Admin)

- **Executive Oversight Dashboard:** Consolidated real-time telemetry on total registered fleet, active rental trips, Gross Merchandise Value (GMV), platform revenue, and pending verification queues.
- **Exclusive Inspection & Live Approval Engine:** Admins hold the sole prerogative to review registered vehicle paperwork, roadworthiness certificates, and push cars live (Approve Live).
- **Dynamic Live Fleet Controls:** Admins retain exclusive toggle buttons in both Admin Dashboard and the live fleet catalogue to take vehicles offline or restore them upon host maintenance requests.
- **Real-Time Telematics Map:** Fleet-wide GPS tracking monitor visualizing active vehicle movements across Kenyan routes (Nairobi, Mombasa, Mara, Naivasha).
- **User Governance:** Ability to activate, deactivate, or suspend accounts, resolve dispute tickets, and enforce marketplace standards.

Pillar 4: Real-Time Availability & Concurrency Trip Locking

- **Instantaneous Concurrency Lock:** When a vehicle booking is confirmed (CONFIRMED, ACCEPTED, or IN_PROGRESS), the vehicle is automatically locked and marked unavailable across all tourist search feeds in real time.
- **Auto-Release Lifecycle:** Upon trip completion or cancellation, the vehicle is automatically released back to the live fleet without manual intervention.

Pillar 5: Security, Compliance & FinTech Governance

- **Kenya Data Protection Act (KDPA) Alignment:** Strict user data sovereignty, pseudonymized telemetry feeds, and encrypted storage of contact numbers and national identity documentation.
- **Automated Audit Trails:** Every vehicle status change, live toggle, payout disburse, and booking mutation produces an immutable audit record.

4. Role-Based Access Control & Workflow Matrix

Feature / Capability	Tourist / Traveler	Fleet Host (Owner)	Platform Admin
Browse & Search Live Vehicles	Full Access	View Only	Full Access
Register Vehicle to Fleet	No Access	Submit (Starts Offline)	Full Access (Override)
Vehicle Approval & Live Toggle	No Access	Strictly Denied (Read-only)	Exclusive Authority
View Other Hosts' Cars / Earnings	No Access	Strictly Blocked (Private)	Full System Oversight
Book Vehicle via M-Pesa STK Push	Full Access	No Access	Admin Booking Support
Live GPS Trip Tracking	Own Trips Only	Own Vehicles Only	Fleet-Wide Live Map
Withdraw Payouts to M-Pesa	Refund Balance Only	Net Rental Revenue (85%)	Disbursement Control

End-to-End Vehicle Lifecycle & Approval Flow

- Step 1: Onboarding** — Host submits vehicle details & photos. Registered as PENDING_APPROVAL & isLive: false.
- Step 2: Admin Inspection** — Admin reviews documents in Mission Control. Admin clicks Approve Live.
- Step 3: Marketplace Publication** — Car status becomes APPROVED & isLive: true. Visible to all travelers.
- Step 4: Reservation & Auto-Lock** — Tourist pays via M-Pesa STK Push. Car automatically locks as On Active Trip.
- Step 5: Trip Completion** — Host marks trip finished. 85% revenue released to host wallet. Car automatically unlocks.

5. Project Deliverables & Implementation Roadmap

Phase	Key Deliverables & Milestones	Duration	Status
Phase 1 Core Architecture	• React 18 / Vite / TypeScript client foundation. • Responsive safari UI & search catalog with 3D models. • Centralized booking store and state synchronization.	Weeks 1–3	COMPLETED
Phase 2 Host & Admin Portals	• Dedicated Fleet Host Dashboard with photo compression. • Admin Mission Control with approval queues. • Enforced Admin-only live push & host read-only status.	Weeks 4–6	COMPLETED
Phase 3 FinTech & Telematics	• Safaricom Daraja 2.0 M-Pesa STK push & B2C payouts. • OpenCV / GPS vehicle tracking modal and telemetry. • Automated concurrency trip locking on confirmation.	Weeks 7–9	COMPLETED
Phase 4 Privacy & Production	• Strict multi-tenant notification isolation (zero leak). • Production build optimization & bundle minification. • Multi-workspace sync and cloud deployment readiness.	Weeks 10–12	COMPLETED & VERIFIED

6. Commercial Model & Executive Sign-Off

Platform Monetization & Revenue Dynamics:

- **Platform Commission:** 15% automatic commission retained on all completed vehicle hires; 10% on bus seat bookings and safari tour packages.
- **Host Payout Structure:** 85% net earnings guaranteed to hosts upon verified trip completion with zero chargeback risk.
- **Value-Added Services:** Premium vehicle verification badges, dedicated safari guide dispatch fees, and corporate invoicing.

Authorized Executive Sign-Off	Date & Formal Acceptance
For M-TRAVEL TOURS (Client / Platform Steering): Name: _____ Title: _____ Signature: _____	Formal Acceptance Date: Date: _____, 2026 Status: [X] Scope Approved [X] Architecture Signed Off Official Seal / Stamp: _____
For MonarchKingz Architecture & Delivery: Lead Solutions Architect: <i>Antigravity Lead Engineer</i> Delivery Status: <i>Codebase Synchronized & Build Passed</i>	Verification Fingerprint: Build Engine: Vite 5.4 / React 18 / TypeScript Production Test Status: 0 Errors (Exit Code 0)